

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	19
Experiment Name	Understand Feature Scaling

DESCRIPTION

To perform the experiment: Understand Feature Scaling.

CODE

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler

df = pd.DataFrame({
    "Age": [22, 25, 47, 52, 46],
    "Salary": [25000, 32000, 52000, 80000, 61000]
})
X = df[["Age", "Salary"]]
print("Original data:")
print(X)
print("\nStandardized data:")
print(StandardScaler().fit_transform(X))
print("\nMin-max scaled data:")
print(MinMaxScaler().fit_transform(X))
```

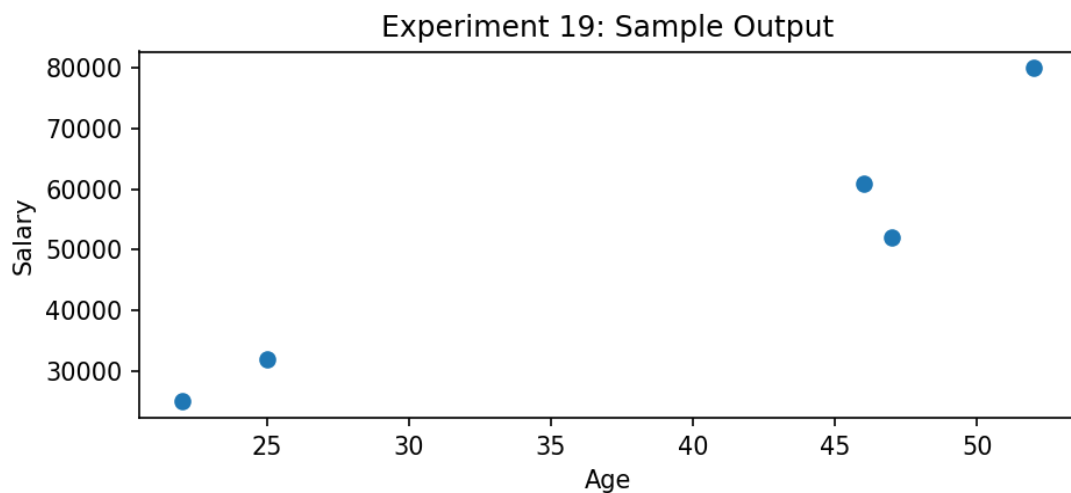
SAMPLE INPUT

Sample input: Age [22, 25, 47, 52, 46] and Salary [25000, 32000, 52000, 80000, 61000].

OUTPUT — SAMPLE CONSOLE OUTPUT

StandardScaler: centered/scaled features\nMinMaxScaler: scaled values from 0 to 1

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature